

Spring Boot Fundamentals

1. What is Spring Boot?

Spring Boot is a framework built on top of Spring that simplifies Java application development. It provides auto-configuration, embedded servers, starter dependencies, and production-ready features.

2. Advantages

- Auto Configuration
- Embedded Tomcat/Jetty
- Starter Dependencies
- Rapid Development
- Production Ready (Actuator)

3. Spring Boot Architecture

Client → Controller → Service → Repository → Database
Controller handles HTTP requests. Service contains business logic. Repository interacts with the database using Spring Data JPA.

4. Common Annotations

@SpringBootApplication - Main application entry.

@RestController - REST API controller.

@GetMapping - Handles GET requests.

@PostMapping - Handles POST requests.

@Service - Business logic layer.

@Repository - Database layer.

@Autowired - Dependency Injection.

5. Dependency Injection

Spring creates and manages objects (beans) inside the IoC Container. Instead of creating objects manually using new, Spring injects required dependencies automatically.

6. Spring Data JPA

Spring Data JPA reduces boilerplate SQL code. Typical repository: UserRepository extends JpaRepository. Methods like save(), findAll(), findById(), deleteById() are automatically available.

7. Spring Security

Provides authentication and authorization. Common features include JWT authentication, role-based access control (RBAC), password encryption using BCrypt, and CSRF protection.

8. REST APIs

HTTP Methods: GET - Read POST - Create PUT - Update DELETE - Delete PATCH - Partial Update

9. application.properties

```
spring.datasource.url=... spring.datasource.username=... spring.datasource.password=...  
server.port=8080
```

10. Best Practices

Use layered architecture, constructor injection, DTOs, validation, exception handling, logging, pagination, and Swagger/OpenAPI documentation.

11. Mini Interview Questions

What is Dependency Injection? Difference between @Component, @Service and @Repository?
What is Auto Configuration? How does Spring Boot start an embedded Tomcat server? What is Spring Security? Difference between Authentication and Authorization?